Question 2

```
clearvars, clc
Rab = readmatrix('q2dat.xlsx')
```

```
Rab = 3×3
    0.9117    0.3221   -0.2552
   -0.2279    0.9131    0.3381
    0.3419   -0.2501    0.9059
```

```
gab = [Rab,[1.8;-0.2;0.05];zeros(1,3),1]
```

```
gab = 4×4
    0.9117    0.3221   -0.2552    1.8000
   -0.2279    0.9131    0.3381   -0.2000
    0.3419   -0.2501    0.9059    0.0500
         0         0         0    1.0000
```

```
x2 = [-1;0;0];
y2 = [0;-1;0];
z2 = cross(x2,y2)
```

```
z2 = 3×1
     0
     0
     1
```

```
Rac = [x2,y2,z2]
```

```
Rac = 3×3
    -1     0     0
     0    -1     0
     0     0     1
```

```
gac = [Rac,[0;-1.2;-0.05];zeros(1,3),1]
```

```
gac = 4×4
   -1.0000         0         0         0
         0   -1.0000         0   -1.2000
         0         0    1.0000   -0.0500
         0         0         0    1.0000
```

Part (a)

MLS eq. (2.24) p. 37:

$$g_{ac} = g_{ab} g_{bc}$$

Thus,

$$g_{bc} = g_{ab}^{-1} g_{ac}$$

```
gbc = gab\gac
```

```
gbc = 4×4
   -0.9117    0.2279    0.3419   -1.4473
   -0.3221   -0.9131   -0.2501   -1.4678
    0.2552   -0.3381    0.9059    0.0306
         0         0         0    1.0000
```

Note that if you type in MATLAB "gbc = gac/gab", you are gettng the wrong answer, as you are calculating $g_{ac}g_{bc}^{-1}$:

```
gbc_wrong = gac/gab
```

```
gbc_wrong = 4×4
   -0.9117    0.2279   -0.3419    1.7037
   -0.3221   -0.9131    0.2501   -0.8154
   -0.2552    0.3381    0.9059    0.4316
         0         0         0    1.0000
```

Part (b)

Using MLS Proposition 2.5 p. 29 to calculate ω and θ.

```
[omega,theta] = omeg_thet(Rab)
```

```
omega = 3×1
   -0.5867
   -0.5956
   -0.5486
theta =
0.5250
```

```
dum1 = rotm2axang(Rab)   % Note: MATLAB equivalent of omeg_thet
```

```
dum1 = 1×4
   -0.5867   -0.5956   -0.5486    0.5250
```

```
shouldbezero = dum1(4) - theta
```

```
shouldbezero =
0
```

```
shouldbezero1times3 = omega'-dum1(1:3)
```

```
shouldbezero1times3 = 1×3
10^-15 ×
   -0.1110   -0.1110   -0.1110
```

```
rad2deg(theta)
```

```
ans =
30.0811
```

Using MLS proposition 2.9 p. 42 to calcualte the exponential coordinates of g_{ab}:

$$e^{\hat{\xi}\theta} = \begin{bmatrix} e^{\hat{\omega}\theta} & [(I - e^{\hat{\omega}\theta})\hat{\omega} + \theta\omega\omega^T]v \\ 0 & 1 \end{bmatrix} = g_{ab} \; .$$

```
AA = (eye(3) - Rab)*hatm_num(omega) + theta*omega*omega'
```

```
AA = 3×3
    0.5094    0.0822   -0.0726
   -0.0656    0.5097    0.0868
    0.0879   -0.0713    0.5084
```

```
v  = AA\gab(1:3,4)
```

```
v = 3×1
```

```
    3.4440
    0.1322
   -0.4784
```

The twist coordinates $\xi \in \mathbb{R}^6$:

```
xi = [v;omega]
```

```
xi = 6×1
    3.4440
    0.1322
   -0.4784
   -0.5867
   -0.5956
   -0.5486
```

Part (c)

According to the definition of the adjoint matrix, MLS p. 55, it is used to transform twists from the body frame to the spatial frame by

$\xi = \mathrm{Ad}_{g_{ab}} \xi^b$,

where ξ and ξ^b are the twist coordinates in the inertial and the body frames, respectively. Therefore

$\xi^b = \mathrm{Ad}_{g_{ab}}^{-1} \xi$.

```
xib = adjmginv(gab)*xi
```

```
xib = 6×1
    3.4440
    0.1322
   -0.4784
   -0.5867
   -0.5956
   -0.5486
```

```
xi - xib
```

```
ans = 6×1
10^-14 ×
   -0.0444
    0.1193
   -0.0389
         0
   -0.0111
         0
```

Part (d)

Since $\omega \neq 0$ and furthermore $||\omega|| = 1$, according to MLS eq. (2.43) p. 47:

$l = q + \lambda\omega$,

where

$q = \widehat{\omega} v$

is the coordinates of a point on the screw axis. The direction of the axis is given by ω.

The pitch of the screw is given by MLS eq. (2.42):

$h = \omega^T v$

and the magnitude by $M = \theta$ (see text just below MLS eq. (2.44)).

```
q = cross(omega,v)
```

```
q = 3×1
    0.3575
   -2.1702
    1.9737
```

```
omega
```

```
omega = 3×1
   -0.5867
   -0.5956
   -0.5486
```

```
hatm_num(omega)*v
```

```
ans = 3×1
    0.3575
   -2.1702
    1.9737
```

```
h = omega'*v
```

```
h =
-1.8370
```

```
Magnitude = theta
```

```
Magnitude =
0.5250
```